Addressing Mechanisms and Heterogeneity in Change with Usable Tools

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Psychological, behavioral, and health processes unfold over time and differ meaningfully across people. In this talk, I present a research program aimed at making dynamic methods more usable for substantive research. I briefly introduce work on continuous-time mediation and on the common and unique latent transition analysis model (CULTA), a model for separating stable and situational influences in alcohol intoxication dynamics. I then focus on MetaVAR, a scalable framework that moves from person-specific dynamic models to valid population-level inference. MetaVAR makes it possible to study average dynamics, between-person heterogeneity, and cross-parameter covariation in a single framework, helping researchers ask not only what changes on average, but how dynamic processes differ across people. Across studies, I treat heterogeneity not as noise to be averaged away, but as a substantive target of explanation, and I close by outlining future directions for building scalable, open, and usable tools for studying dynamic processes across people, settings, and timescales.